Chenghan Wu 武承翰

APPLIED MATHEMATICS / ARTIFICIAL INTELLIGENCE

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RESEARCH DIRECTION

I study numerical foundation models, extending large language model training and in-context learning to numerical problems across scientific disciplines. My work spans in-context operator networks, inference-time harnesses, and automated scientific research.

Broader interests: large language models, digital and artificial life, and machine aesthetics.

CURRENT APPOINTMENT

Research Assistant, Department of Mathematics, National University of Singapore Sep 2026–Present

Scientific Computing and Intelligence Group (Scaling Group); advisor: Prof. Liu Yang

- Develop numerical foundation models that acquire system-specific predictive knowledge from context and generalize across scientific disciplines without task-specific retraining.
- Investigate automated scientific research with language-model agents for data curation, experiment design, model development, evaluation, and iterative self-improvement.
- Contribute to shared research infrastructure for reproducible model development, training, and evaluation across the group.

SELECTED RESEARCH

UNICON **Foundation Model of Numerical Intelligence** 2026–Present

- Designed and developed a frozen foundation model trained across heterogeneous graph-based systems spanning hydrology, traffic, power, weather, ocean, land, soil, solar resources, and human mobility.
- Demonstrated transfer to disciplines absent from training while approaching specialist performance; combined UNICON with LLM-agent-driven contextual ensemble learning to surpass specialist baselines on an unseen discipline.
- Develop inference-time harnesses around UNICON in which language-model agents construct, test, and refine numerical reasoning strategies without updating model parameters.

GICON **Graph In-Context Operator Networks** 2025–2026

- Designed and implemented graph message passing and example-aware positional encoding for geometric and variable-cardinality generalization.
- Established controlled comparisons with single-operator learning under matched data and training budgets; demonstrated cross-region transfer and robust scaling from few examples to 100 examples at inference on real-world air-quality forecasting.

SELECTED RESEARCH (CONTINUED)

CHOP **Chain of Operators** 2026

- Co-developed an interpretable inference-time harness that adapts frozen scientific foundation models to out-of-distribution tasks through explicit mathematical transformations and compositional model calls, without parameter updates.
- Evaluated reusable operator chains across PDE benchmarks and real-world spatiotemporal forecasting, establishing a programmable route toward agentic scientific computing.

ICON-Core **Open Research Infrastructure for ICON** 2025–2026

- Contributed to the group's open-source core infrastructure for scaling in-context operator learning across heterogeneous numerical systems, supporting standardized multi-dataset integration, shared training and evaluation pipelines, and reproducible experiment management.

SELECTED PUBLICATIONS

A Foundation Model of Numerical Intelligence with Cross-Disciplinary Generalization
Chenghan Wu, Zongmin Yu, Liu Yang. *arXiv preprint, 2026.* arXiv:2607.28432.

Graph In-Context Operator Networks for Generalizable Spatiotemporal Prediction
Chenghan Wu, Zongmin Yu, Boai Sun, Liu Yang. *2nd ICML Workshop on Foundation Models for Structured Data, 2026.* arXiv:2603.12725.

Chain of Operators: An Inference-Time Harness for In-Context Operator Learning
Minghui Yang, **Chenghan Wu**, Ling Guo, Liu Yang. *arXiv preprint, 2026.* arXiv:2606.12318.

EDUCATION

National University of Singapore 2024–2026

M.Sc. in Statistics

Shanghai Jiao Tong University 2020–2024

B.Eng. in Naval Architecture and Ocean Engineering; B.Sc. in Mathematics and Applied Mathematics (double degree)

HONORS

Undergraduate Merit Scholarship, Shanghai Jiao Tong University	2022–2023
Undergraduate Merit Scholarship, Shanghai Jiao Tong University	2021–2022
Undergraduate Merit Scholarship, Shanghai Jiao Tong University	2020–2021